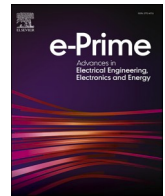




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Glaucoma detection from retinal fundus images using graph convolution based multi-task model[☆]

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ABSTRACT

Glaucoma is an abnormality in the eye condition that, if not treated within a safe time limit, can result in visual loss. Glaucoma diagnosis requires professionals to identify minor structural changes in the structure of the optic disc and optic cup from retinal fundus images in a short period. Deep learning algorithms have been employed effectively in the segmentation of biomedical images and the identification of diseases. To accomplish good generalization, model training requires comprehensive annotations, which is a difficult task. The intended objective of the present study is to come up with and train a distinctive multi-task deep learning model for automated fundus image segmentation and classification. The multi-task model learns for the segmentation task of Optic Disc (OD) and Optic Cup (OC) and the classification task for accurate glaucoma detection using both structural and image-based features. The multi-task model proposed a modified U-net architecture in which Mobile-Netv2 is used in the encoder part, Graph Convolution Network (GCN) is used in the decoder part, and an attention module (AM) is used to locate the region of interest (ROI) for better feature extraction. The implementation of this model is done using three fundus image datasets such as ORIGA, REFUGE, and DRISTI-GS. The performance of the proposed multi-task model is compared with some existing methods. It shows maximum accuracy of 97.43 % and AUROC of 0.985 for the glaucoma detection task and high-quality OD and OC segmented images with dice coefficient of 97.95 % and 96.11 % respectively for the segmentation task.

1. Introduction

Glaucoma is a major eye disease caused by damaging of optic nerves due to a sudden rise in intra-ocular pressure (IOP) inside the eye. The optic nerves are connected from the eye to the brain for sending visual signals, which are interpreted as images. When the excess fluid inside the eye does not drain out, causes an increase in eye pressure which in terms of damage to the retinal nerve fiber (RNF). The deterioration of the optic nerve causes an anatomical change in the size of OD and OC [1]. The discrepancy in the size of the OC concerning the OD indicates the possibility of glaucoma, a degenerative eye disease. So early detection of glaucoma is highly needed to forbid permanent vision loss.

Open-angle and closed-angle glaucoma are the two different types of glaucoma generally. In closed-angle glaucoma, all the drainage angles are closed and the fluid inside cannot be drained out. So, a sudden increase in IOP occurs and the patient has some symptoms like redness in the eye, vomiting, decreased vision, and pain inside the eye [2].

Compared to closed-angle, open-angle glaucoma is more dangerous because of no early symptoms. In that case, some drainage channels are open to drain out the excess fluid and the patient is unaware of his condition until some noticeable vision loss. Fig. 1 shows (a) a sample of the fundus image, (b) the boundary of OD and OC, and (c) ISNT regions from the publicly available dataset. The visual fields for normal and glaucomatous eyes are also shown in Fig. 1(d) and (e).

The clinical measurement of glaucoma includes a series of tests such as tonometry, ophthalmoscopy, perimetry, pachymetry, gonioscopy, etc. [3]. The increase in pressure inside the eye is measured in a tonometry test. If the pressure is >20 mmHg, then there is an indication of a glaucomatous eye. The unusual changes in the anatomy of the optic nerve like color and shape are examined by ophthalmoscopy. Then, another test called perimetry which determines the effects of vision due to glaucoma. Due to increased IOP, the thickness of retinal layers is getting affected and that can be examined by pachymetry. Another clinical test called gonioscopy determines the condition of the drainage

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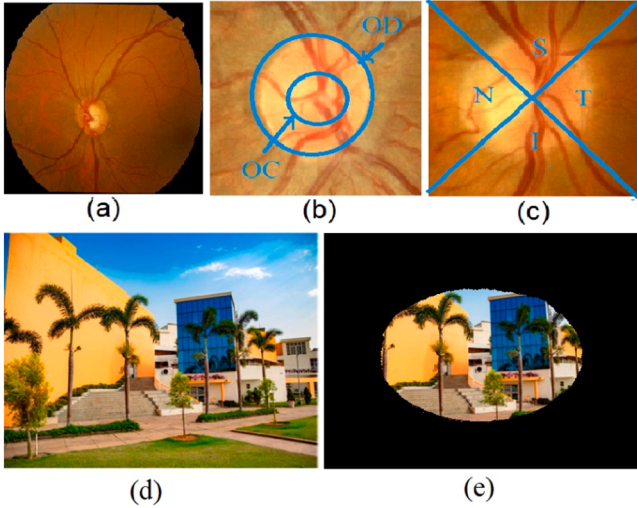


Fig. 1. Sample of fundus images showing Optic Disc and Optic Cup from REFUGE dataset [60].

angle whether it is open or narrow. The risk of glaucoma is very high in case of a narrow drainage angle. However, these clinical tests are time-consuming, expensive, and always need trained and professional ophthalmologists [4]. So, a reliable and automatic glaucoma detection method is best suited using computer-aided diagnosis approaches.

Retinal fundus imaging is a cost-effective and robust method that provides an analysis of anatomical changes obtained in the fundus images [5]. Several computer-aided tools have come up with image-processing approaches for early diagnosis of glaucoma. Optic nerve head (ONH) is one of the effective tools which includes OD and OC region analysis. Cup to disc ratio (CDR) is another popular feature indicator for glaucoma diagnosis [6]. CDR is the ratio between the area of OC and the area of OD. The analysis of the Neuro Retinal Rim (NRR), a portion between OC and OD, is also an indicator for glaucoma assessment. ISNT rule is also used for glaucoma detection by analyzing the region of inferior, superior, nasal, and temporal rim areas [7]. An important part of these detection methods is the accurate segmentation of OD and OC using Histogram of Gradient (HOG) [62].

Manually annotated segmentation takes much time, is less reliable, and also requires expertise. Deep convolutional networks have been successfully showcased in the field of medical imaging to aid in creating learners and directly extracting patterns from images. Like efficient and alternative support to ophthalmology, deep learning has played an important role in automatic glaucoma detection [8]. Deep learning (DL) models require a vast quantity of resources to be properly trained from starting point to completion, otherwise the most appropriate weights can not be updated. This problem leads to a reduction in the accuracy of that model which gives rise to the solution called transfer learning [9]. It is another popular strategy for glaucoma detection using a pre-trained model with a small dataset. For improving generalization performance, multitask learning (MTL) [10] leverages domain-specific data contained in the training models of some related tasks.

Many important real-world objects have a complex structure such as images are considered as a special case of graphs in which the nodes lie on regular 2D lattices [11]. Every individual pixel of an image is portrayed as a node when converted in the form of a graph. Deep learning regarding images in graph formatted information has the benefit of encoding complicated pairwise relationships by learning the most informative representation. Another advantage is that the framework and logical content of original images can be used to include knowledge specific to the domain for getting precise interactions with data, which is known as a graph neural network (GNN). Convolution operation in GNN, Graph Convolution Network (GCN) has been used for biomedical

image segmentation through boundary-aware input-dependent GCN approach [12]. The proposed Laplacian method used two attention modules for efficient segmentation.

In present scenario, the challenges in this research fields occur due to the unavailability of huge number of datasets. Additionally, using unsuitable datasets can also lead to researchers creating their own private ground-truths. When the same segmentation method is used for both the optic disc and the optic cup, the results for the optic cup are often less satisfactory. This is because many methods do not consider the relationships between different parts of the retina. The accuracy of distinguishing different parts of the retina largely depends on the extracted features, which in turn depend on several parameters. However, most methods only take into account a few of these parameters, and large datasets are necessary for optimal network training. Unfortunately, such extensive datasets are not easily available for glaucoma patients, as the data collection and training processes are time-consuming.

In this research, a modified U-Net model is proposed for a multi-task approach which includes OD segmentation, OC segmentation, and glaucoma classification using retinal fundus images. Being motivated by Kanadath et al. [13] Mobile-NetV2, a pre-trained model is used in the contracting path of the U-Net model which solves the huge data requirement problem with fewer parameters for training. In the decoder part of the U-Net model, we have added residual learning for GCNs [14] along with the Deep-CNN model which extracts important spatial information for OD and OC boundary location in Euclidean space. An Attention module (AM) [15] is used in the skip connections between the encoder and decoder part to locate the ROI of the fundus image with better feature extraction. The classification task also consists of two steps. (1) The features obtained from the encoder part are used to classify a fully connected network. (2) A linear classifier classified retinal images based on CDR computed from segmented OD and OC.

The key contributions of this work are outlined below.

- We propose a modified multi-task approach for OD and OC segmentation and image classification using U-Net as backbone architecture.
- We modify the basic U-Net model by introducing Mobile-NetV2 in the encoder part which solves the problem of the requirement of huge data for training, and Residual-Graph Convolution Network (RES-GCN) used in the decoder part which extracts more spatial features for segmentation task.
- We implement an Attention Module (AM) in the skip connections between the encoder and decoder part of the proposed modified U-Net model which improved the segmentation task by locating the Region of Interest (ROI) of the input fundus image.
- Our experimental results outperform compared with state of art methods for OD and OC segmentation with Dice Coefficient of 97.95 % and 96.11 % respectively, and glaucoma detection task with an accuracy of 97.43 %.

The subsequent portions of this paper are formatted as follows. Section 2 describes some existing works related to segmentation and glaucoma classification. The proposed methodology along with details about the datasets used are explained in section 3. It also contains the architecture of the proposed work in detail individually. The implementation results and comparative state of art methods are explained in section 4. The last section 5 concludes the whole work with future expectations.

2. Related works

The segmentation of OD and OC is critical for extracting features such as CDR for early and automated glaucoma detection by implementing DL methods.

A. Sevastopolsky [16] alimented a modified version based on the previous U-Net model for OD and OC segmentation, in which the image

is routed via the contraction and expansion routes of the network with up sampling layers that increase image dimensions. The model yields high-quality segmented OD and OC in the quickest possible period. J. Zilley et al. [17] created an ensemble model based on a CNN for OD and OC segmentation. By determining the most informative points, the entropy sampling technique was utilized to construct a learning architecture for convolutional filters. When a small dataset is provided, the suggested study is beneficial. A DL-based model called M-Net was used in [18]. Using polar transformation, original fundus pictures are converted to polar coordinates and put into M-net to generate probability maps for disc and cup regions using the ORIGA dataset. The researchers [19] put five alternative ImageNet-trained networks to the test as glaucoma diagnosis systems. Analyzing these models on publicly accessible datasets yielded high accuracy, sensitivity, and specificity. In [41] OD is segmented using non-convex rank approximation by separating the background from the low-rank foreground of the fundus image and for specular removal [61]. Multi-task DL models have also been presented in healthcare applications such as image segmentation and classification of diseases. Mo-jab et al. [20] suggested a multi-task framework for glaucoma identification that included two modules for both OD and OC segmentation and glaucoma disease prediction with an F-score of 90.1. The researchers failed to compensate for the disparity in segmentation and prediction task distributions. Chelaramani et al. [21] presented a unique MTL-based instructor ensemble knowledge distillation approach. The suggested technique necessitates a dataset including a wide range of ocular diseases, which may be challenging to collect in practice. A novel segmentation approach has been developed that integrates a pretrained ResNet-34 structure as an encoder with conventional U-Net as a decoder [22]. The RIGA dataset was used to train this network. Segmentation was accomplished out in [23] using a modern U-net architecture. The cropped and scaled-down images were used for loading into the updated U-net neural network with increasing convolutional layers and decreasing parameters for OC segmentation. The identical procedure was used for OD segmentation, except for cropping. The approach was tested using the DRISHTI and RIM-ONE datasets.

Deep learning applied to graphs has given rise to a new out-of-domain topic known as the graph-based DL model, which attempts to learn functional representations of graphs from input to output. A. Joshi et al. [24] developed a three-layer graph-based deep framework for OD and OC segmentation with an upgraded fusion approach. CNN encoder-decoder architectures, supplementary graph networks, and matching via fusion-based rules are researched for connecting local and global data. According to this method, the dice similarity coefficient for the DRISHTI-GS dataset of 0.97 and the RIM-ONE dataset of 0.96 was obtained. Similarly, the intersection of unions utilizing those datasets for OD measurement was 0.96 and 0.93, respectively. Similarly, the intersection of union for OD measurement using the DRISHTI-GS and RIM-ONE datasets was 0.96 and 0.93, respectively. DSC and IOU for OC segmentation in DRISHTI-GS data were 0.93 and 0.90, respectively, and 0.83 and 0.82 in RIM-ONE data. Meng et al. [15] employed a multilevel fusion network rather than a pixel-wise dense prediction to reconstruct the coordinates of OD and OC borders. A CNN, an attention-reinforced module, and a GCN are combined in this model. To extract more effective information, the attention module acts as an intermediary between the CNN encoder and the GCN decoder. This model, like prior work [25] by the same authors, extracts feature correlations between GCN layers.

In Thisara et al. [51], three CNN architectures are recommended for attention U-Net models in segmenting fundus images: Inception-V3, Visual Geometry Group 19 (VGG19), and Residual Neural Network 50 (ResNet50). To enhance accuracy and prevent overfitting, the researchers explored three distinct CNN architectures (Inception-V3, VGG19, and ResNet50) and employed various data pre-processing and augmentation techniques [52]. Data pre-processing methods like dilation and Contrast Limited Adaptive Histogram Equalization (CLAHE) were utilized. The researchers used performance metrics from image

classification and employed modified architectures for the Inception-V3, VGG19, and ResNet50 models [53]. To identify the best training parameters, the rectified linear activation function (ReLU) was used for the internal layers.

Traditionally, glaucoma diagnosis relies on methods like tonometry and pachymetry, which can be time-consuming and prone to errors. The research [42] proposes an AI-based computer-assisted diagnosis system that uses machine learning to classify retinal images as “healthy” or “infected”. The system employs a new set of hybrid features selected from 36 initial features using the Emperor Penguin Optimization and Bacterial Foraging Optimization algorithms. The system’s performance is evaluated using seven machine learning classifiers, achieving a high specificity of 0.9940, sensitivity of 0.9347, and accuracy of 96.55 %. This tool aims to assist medical professionals in making accurate glaucoma diagnoses, potentially alleviating their workload. With glaucoma cases expected to rise significantly, the study [43] uses retinal fundus images to extract 36 features from both public and private datasets. Six Machine Learning (ML) models are trained using the most influential features selected by GSOA. The method achieved a 95.36 % accuracy rate, evaluated through multiple cross-validation techniques. The segmentation process is crucial for applications in biometric authentication, computer-assisted surgeries, and disease detection, such as diabetes and heart disease. The research introduces [44] a method for automatically extracting blood vessels from colored retinal images using pixel-based feature extraction and ML techniques. The method utilizes five feature groups and trains four ML classifiers: K-Nearest Neighbor (KNN), Support Vector Machine (SVM), Naive Bayes, and Decision Tree. Tested on the DRIVE dataset, the approach achieved high accuracy rates, with Decision Tree yielding the highest at 0.9624. The proposed method demonstrates comparable performance to other leading techniques and offers an effective automated solution for retinal vessel segmentation. Another study proposes three CNN models [45]: one built from scratch, an ensemble of five top-performing networks, and a CNN-LSTM hybrid model. These models were tested on three Kaggle datasets, achieving impressive performance metrics, including a maximum accuracy of 0.9545, a maximum F1 score of 0.9685, and a maximum AUC score of 0.9769. The proposed networks demonstrated superior performance compared to existing models, indicating their potential in effectively diagnosing retinal issues in diabetic patients, thereby aiding in timely treatment and preventing vision loss.

In summary Table 1 describes, the discussed methodologies differ in computational requirements compared to the baseline multi-task approach. Currently, there are two prevalent methods for diagnosing glaucoma using deep neural networks. The first involves segmenting the optic disc and cup, followed by morphological analysis of the structures. The second approach directly addresses diagnosis as an image classification task. The segmentation method benefits from pixel-level labels with precise morphological information for training. However, while the classification approach lacks this detailed training feedback, it can use image-level labels to uncover additional non-morphological cues that are also relevant for diagnosis. Instead of training separate models for segmentation and classification, a multi-task model consolidates these tasks, reducing computational resources and training time for large-scale deployment in clinical settings.

3. Materials and methodology

3.1. Data resources

The summarized detail about each dataset is described in Table 2.

3.2. Proposed multi-task deep learning (MDL) model

This research work proposes a modified architecture of a special type of Autoencoder called U-Net as a multi-tasking model as seen in Fig. 2. This is an encoder-decoder convolutional network with a Mobile-NetV2

Table 1

List of literature on glaucoma classification.

Ref	Year	Methods	Database	Strengths	Limitations
[48]	2021	Attention-guided stereo ensemble network	Stereo glaucoma image dataset	Utilizes extensive information from stereo image pairs to ensure reliability and robustness and also incorporates attention mechanisms for regions of interest.	The performance of the method can be improved by increasing the batch size during training which is not feasible in the study due to system complexity.
[49]	2022	Deep Neural Network	ORIGA-light	Enhanced glaucoma detection using deep learning techniques.	The specifics of the deep learning model is not extensively discussed, and specific performance metrics may be lacking. Scalability considerations are not addressed.
[50]	2022	EfficientDet-D0 with EfficientNet-B0	ORIGA	EfficientNet-B0 for key point extraction to improve glaucoma recognition performance while reducing both training and execution time.	Classification is based on very less features. More robust feature selection methods can be implemented in this model to extend this work to other eye diseases.
[46]	2023	Deep Neural Network	DRISTI-GS, ORIGA	Enhanced classification of retinal fundus images using deep learning techniques.	The proposed model did not tackle the problem of the blurred boundary between the optic disc (OD) and optic cup (OC).
[47]	2023	Rim-Net	DRISTI-GS, Internal test-set	Automatic detection of the optic disc rim using a deep neural network pipeline.	The details of the deep neural network architecture are not extensively discussed. Specific performance metrics may be lacking, and scalability considerations are not addressed.

structure in the contracting path and additional skip connections between encoder and decoder depths, allowing the decoder to retrieve fine-grained information via successive up sampling. The attention module is used in the skip connections to locate the ROI boundaries. We use a modified version of the decoder network for efficient segmentation. Graph Convolution Network (GCN) is introduced in the expansion path after up sampling the direct features through skip connections. The features are then concatenated, convoluted, and added to the expansion

Table 2

Datasets used for experimental analysis.

Dataset	Total images	Normal	Glaucoma	Training (70 %)	Testing (30 %)
ORIGA [28]	650	482	168	455	195
DRISTI-GS [26]	101	31	70	71	30
REFUGE [27]	1200	1080	120	840	360
Total	1951	1593	358	1336	585

group. This architecture is used in our multi-task model for two segmentation tasks (OD and OC), and one classification task (glaucoma diagnosis).

3.2.1. OD and OC segmentation task

Following the common decoder for each segmentation task, the GCN and convolutional layers are used to generate the OD and OC masks. Similar to the basic expansion path, the OD and OC segmentations are refined using residual GCN (Res-GCN) that preserves the primary linked element in the prediction model to reduce any projected noise around these elliptical sections.

3.2.2. Glaucoma detection task

The glaucoma detection (classification) work is divided into two steps:

- A prediction was computed by a fully connected layer that was diverged after the U-Net encoder and attention module.
- A linear classifier is used to construct a second prediction, with the vertical CDR obtained from the OD and OC segmentation as input.

$$CDR = \frac{\text{Area of OC}}{\text{Area of OD}} \quad (1)$$

3.3. Network architecture

The detailed architecture is shown in Fig. 3 and is described in depth.

3.3.1. Encoder (Mobile Net)

Convolutional neural networks optimized for use with mobile devices are used in Mobile Net models. According to Howard et al., [29] model, MobileNetV1 features are exclusive features called depth-wise independent filters. Because depth-wise separable convolutions need fewer calculations, the deep network is computationally less expensive than a regular convolution operation. MobileNetV2 [13] is a variant in the Mobile Net family that is faster than MobileNetV1. The model offers two advantages over MobileNetV1. They have linear bottleneck layers that are linked together via alternate links.

The amount of data passing over the network can be reduced by using bottleneck layers. There are two types of bottleneck layers in MobileNetV2: stride 1 and stride 2. Every bottleneck layer is composed of three layers in turn. The first layer is an 1×1 convolutional layer, followed by a ReLU6 operation. The rectified linear unit (ReLU-6) is an enhanced version of the ReLU.

Let's consider the following assumptions.

F = input fundus image of size $r \times c \times h$ = number of channels f = number of filters with a size of $r_1 \times c_1$

(1) and (2) provide the ReLU function and the ReLU6 function for the input r , respectively.

$$\text{ReLU}(r) = \max(r, 0) \quad (2)$$

$$\text{ReLU}_6(r) = \min(\max(r, 0), 6) \quad (3)$$

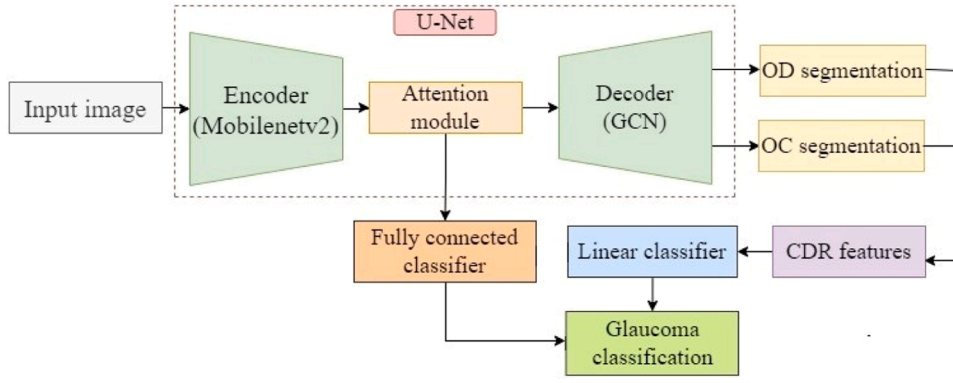


Fig. 2. Block diagram of proposed multi-task model.

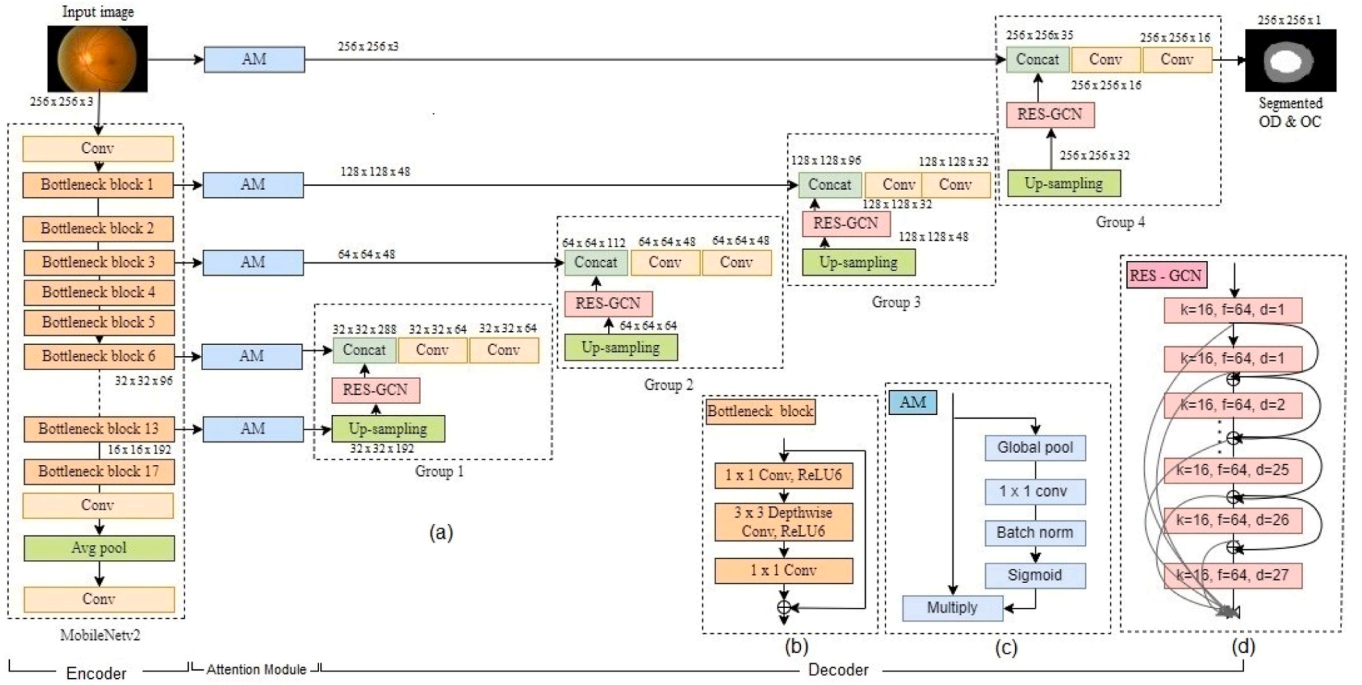


Fig. 3. Details internal architecture of proposed modified U-net model.

The preceding layer's data is collected by the bottleneck layer, thereby increasing the number of data channels. The second layer has 3×3 filters that perform depth wise separable convolution. The retrieved features are forwarded to the third layer. As illustrated in Fig. 3(b), the third layer is a projection layer, which is an 11 convolutional layer without a ReLU operation to minimize the number of channels.

The next important MobileNetV2 feature is a skip connection or a shortcut connection, which is a residual connection between bottleneck levels. The likelihood of vanishing gradients in a deep model increases as the number of layers' increases. This problem can be solved by residual connection or skip connection. Skip connections give the gradient an alternate path. When back propagation occurs, these pathways will transmit the gradient across the network and function to update the weights of previous layers as shown in Fig. 3(b). There are multiple levels to the MobileNetV2 architecture. The first layer is a convolution layer of 32 filters, followed by 17 bottleneck levels. The MobileNetV2 architecture is thoroughly documented in [13]. Fig. 3 depicts the fundamental structure of the MobileNetV2 model, which is used as an encoder network in our proposed model.

3.3.2. Attention module (AM)

We introduce an Attention Module (AM) [15] that easily combines global context information to improve the output characteristics of each encoder level (row). AM is made up of five attention blocks, as shown in Fig. 3(c).

Each of these uses global channel average pooling to collect global context across multiple channels and the computation occurs to find an attention vector that guides the feature learning process. The process occurs through a convolution layer kernel size is 1×1 with stride 1, followed by batch Normalization layer, and sigmoid activation function.

3.3.3. Decoder (GCN)

An image can be represented by a graph of $\mathcal{H}$ of the tuple $\mathcal{H} = (\mathbf{V}, \mathbf{E})$, where the spontaneous vertices set is $\mathbf{V}$ and the connection between the vertices $\mathbf{v} \in \mathbf{V}$ is indicated by $\mathbf{E}$, which is the set of edges. If $e_{ij} \in \mathbf{E}$, then the vertices $\mathbf{v}_i$ and $\mathbf{v}_j$ are linked by an edge e_{ij} .

Vertices are represented by GCNs with a feature vector $\mathbf{g}_v \in \mathbb{R}^d$, which is associated with each $\mathbf{v}$ and d is the dimension of the features. As a result, the graph $\mathcal{H}$ may be represented as a whole concatenation of the features of all the unstructured vertices, i.e. $\mathbf{g}_{\mathcal{H}} = [\mathbf{g}_{v_1}, \mathbf{g}_{v_2}, \dots]$.

$\mathbf{g}_{v_n} \in \mathbb{R}^{n \times d}$, where the cardinality of set $\mathbf{V}$ is represented by n . A generic graph convolution process $\mathcal{G}$ at the l -th layer can be expressed as the techniques for gathering and updating is shown below.

$$\mathcal{H}_{l+1} = \mathcal{G}(\mathcal{H}_l, \mathcal{W}_l) = \text{Update}\left(\text{Aggregate}(\mathcal{H}_l, \mathcal{W}_l^{\text{aggregate}}), \mathcal{W}_l^{\text{update}}\right) \quad (4)$$

$\mathcal{H}_l = (\mathbf{V}_l, \mathbf{E}_l)$ and $\mathcal{H}_{l+1} = (\mathbf{V}_{l+1}, \mathbf{E}_{l+1})$ are the input and output graphs at the l -th, respectively $\mathcal{W}_l^{\text{aggregate}}$ and $\mathcal{W}_l^{\text{update}}$ are the essential components of weights which can be learned in the process of aggregation and update functions respectively.

The underlying mapping $\mathcal{G}$ is learned in the original graph learning paradigm, in which a graph is taken as input and returns a novel representation of the graphs in Eq. (4). In this paper, we adopted a residual learning approach on graphs for a fundamental mapping $\mathcal{H}$ by adapting another mapping $\mathcal{G}$. $\mathcal{H}_{l+1}$ is obtained by vertex-wise addition after $\mathcal{H}_l$ has been converted by $\mathcal{G}$. $\mathcal{G}$ is the residual mapping in which a graph is taken as an input for learning and produces a representation of a graph with the residual function $(\mathcal{H}_{l+1})^{\text{res}}$ for the following layer. The collection of parameters is represented by $\mathcal{W}_l$ that can be learned at layer l . In this research, the residual graph-based convolution network is called RES-GCN.

$$\mathcal{H}_{l+1} = \mathcal{U}(\mathcal{H}_l, \mathcal{W}_l) = \mathcal{G}(\mathcal{H}_l, \mathcal{W}_l) + \mathcal{H}_l = (\mathcal{H}_{l+1})^{\text{res}} + \mathcal{H}_l \quad (5)$$

The outputs of the attention module are provided to the decoder which decodes them using ResGCN blocks [14] at various levels. By using graph sampling, the borders of the object can regress from sparse to dense in the decoder module. Each ResGCN block is made up of 28 edge convolution layers [30] with k -NN in dilated form and residual connections of graphs for all layers. The number of vertices is up sampled from 16 to 256 in the output features of ResGCN blocks. In the graph vertices up samplings, the vector length of 32 is represented for each vertex. The RES-GCN blocks are concatenated, convoluted, and passed to the next group. In Fig. 3(d) the architectures of RES-GCN are depicted, where k is the number of nearest neighbors, f is the number of filters used, and d is the dilation rate. In this model when the network depth increases, the dilation rate (d) of dilated k -NN also increases linearly.

3.4. Performance measures

A ten-fold cross-validation strategy is employed to test the classifiers using the extracted features. In this approach, the entire dataset is divided into ten approximately equal parts. The process begins by training the model on the first nine parts of the data and testing it on the remaining part. This procedure is repeated nine more times, each time using a different set of training and testing data. For each iteration, the accuracy, Positive Predictive Accuracy (PPA), sensitivity, and specificity are calculated. The average values of these metrics across all ten folds provide the overall accuracy, PPA, sensitivity, and specificity.

3.5. Loss function

3.5.1. Segmentation loss

Both the OD and OC segmentation tasks employ a binary cross-entropy loss (BCE) that is averaged across each pixel p of the segmentation maps:

$$\mathbb{L}_{\mathcal{Y}, \mathcal{Y}} = -\frac{1}{N_{\text{pix}}} \sum_{p=1}^{N_{\text{pix}}} \mathcal{Y}_p \log(\mathcal{Y}_p) + (1 - \mathcal{Y}_p) \log(1 - \mathcal{Y}_p) \quad (6)$$

where $\mathcal{Y}$, $\mathcal{Y}$ and N_{pix} are the prediction, ground truth, and number of pixels respectively.

3.5.2. Classification loss

A focal loss [31] is employed in the glaucoma classification job to better address the imbalance between positive and negative data.

$$\mathbb{L}_{\text{Focal}}(\mathcal{Y}, \mathcal{Y}) = (1 - \mathcal{Y}_t)^{\gamma} \log(\mathcal{Y}_t) \quad (7)$$

With

$$\mathcal{Y}_t = \begin{cases} \mathcal{Y} & \text{if } Y = 1 \\ 1 - \mathcal{Y} & \text{otherwise} \end{cases}$$

To provide more weight to uncertain classifications, the typical binary cross-entropy term in this loss is increased by a classification uncertainty term $(1 - \mathcal{Y}_t)$ for less countable classes. The hyper-parameter is set as 2 for our implementation.

3.5.3. Multi-task loss

The overall loss of the proposed multi-task model such as segmentation and classification are as follows.

$$\text{Overall - loss, } \mathbb{L} = a \mathbb{L}_{\mathcal{Y}, \mathcal{Y}} + b \mathbb{L}_{\text{Focal}}(\mathcal{Y}, \mathcal{Y}) \quad (8)$$

The standard Multi-Task Learning (MTL) optimization framework using a combined loss [10] can be described as follows:

$$\mathbb{L}(\mathcal{W}_k, \mathcal{G}_k) = \sum_{t=1}^{\mathcal{M}} C^{(t)} \cdot \mathbb{L}^{(t)}(\mathcal{W}_k, \mathcal{G}_k) \quad (9)$$

where $\mathbb{L}^{(t)}$ is the loss function associated with the t^{th} task out of $\mathcal{M}$ tasks,

$\mathcal{W}_k$ represents the shared parameters, and $\mathcal{G}_k$ is the data sample at iteration k . The $C^{(t)}$ are task-specific weighting coefficients, which we assume to be uniform, i.e., $C^{(t)} = 1$. If $g^{(t)}$ denotes the derivative of $\mathbb{L}^{(t)}$ with respect to the shared parameters $\mathcal{W}$, the update rule for $\mathcal{W}$ at step $k + 1$ using stochastic gradient descent is:

$$\mathcal{W}_{k+1} = \mathcal{W}_k - \eta_k \cdot \sum_{t=1}^{\mathcal{M}} g^{(t)}(\mathcal{W}_k, \mathcal{G}_k) \quad (10)$$

Where η_k is the learning rate.

3.6. Model implementation

The proposed model is implemented using the Google CoLaboratory, a free cloud service for sharing deep learning research. The Google CoLab offers a programming environment with powerful graphics processing (GPU) and deep learning libraries based on Tensorflow. This makes it possible to run a powerful deep-learning network quickly without a personal GPU. The fundus image datasets are uploaded in Google Drive which is connected to CoLaboratory's environment. The fundus images in the dataset are divided into training (70 %) and testing (30 %). Adam optimizer is used with a learning rate of 0.001 and it will be halved for every gradient decent step. By taking 300 epochs the model is used for all three datasets individually.

4. Results and discussion

4.1. Segmentation task

The segmentation results are verified according to the relations between ground truth and the predicted area depicted in Fig. 4. Based on that measurement, the evaluation parameters are calculated such as the Dice co-efficient, and Jaccard as follows.

$$\text{Dice - coefficient (DC)} = \frac{2 \times TP}{((2 \times TP) + FP + FN)} \quad (11)$$

$$\text{Jaccard (Jc)} = \frac{TP}{TP + FP + FN} \quad (12)$$

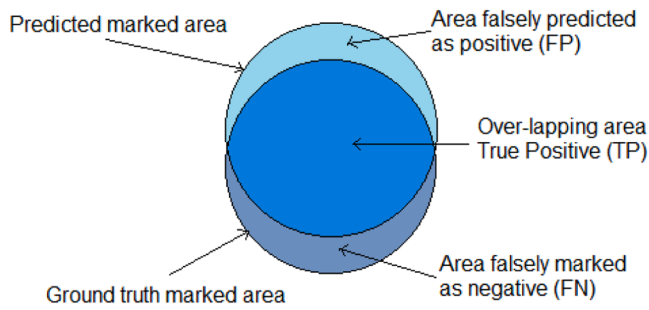


Fig. 4. Segmentation relations between ground truth marked are and predicted marked area.

The proposed technique is implemented in this study using three datasets: ORIGA, DRISTI-GS, and REFUGE. The segmented OD and OC with ground truth are shown in Fig. 6 and the jointly segmented OD and OC binary image is shown in Fig. 5. Table 3 shows the evaluation parameters which are calculated by following the above formula. For OD segmentation using the ORIGA dataset, this research shows maximum output such as dice and Jaccard coefficient of 97.95 % and 95.58 % respectively, and for OC segmentation the dice coefficient is 96.11 % and Jaccard of 94.95 %. For the other two datasets also, the proposed method shows appreciable results. Using the DRISTI-GS dataset the proposed method shows the dice coefficient and Jaccard of 96.56 % and 91.27 % for OD segmentation and 94.11 % and 93.14 % for OC segmentation respectively. Similarly, using the REFUGE dataset this method shows the dice coefficient and Jaccard of 96.59 % and 92.87 % for OD segmentation and 95.23 % and 92.95 % for OC segmentation respectively.

The results of the segmentation task are compared with some existing methods described in Table 4. It is evident that, compared to other related studies that have used segmentation techniques, our approach achieves higher accuracy. Notably, our model performs best with the ORIGA dataset. Additionally, we have utilized three different fundus image datasets, rather than relying on a single dataset, to demonstrate the generalizability of the proposed framework.

4.2. Classification task

The proposed multi-task model is implemented for fundus image classification using both structural and image-based features depicted in Fig. 2. Firstly, the image-based features are obtained from the output of the encoder part i.e. Mobile-NetV2 of the proposed model. These image-

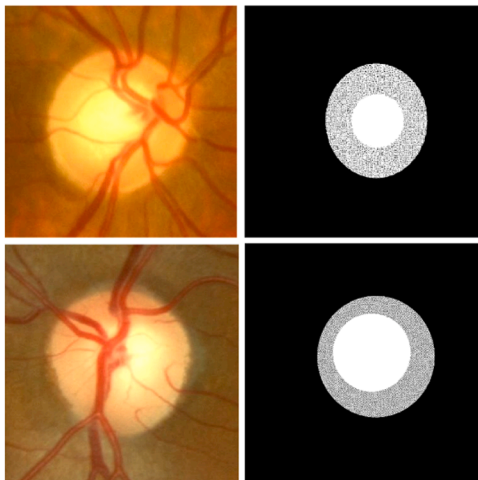


Fig. 5. Cropped fundus images (left), cropped binary OD and OC segmented images (right).

based features are taken as input to the fully connected classifier for glaucoma prediction. Secondly, the obtained structural features such as OD area, OC area, and calculated CDR features are taken as input to a linear classifier to distinguish the fundus images as normal or glaucoma. The output confusion matrices obtained from the classification task are shown in Fig. 7 using three different datasets. The results of the classification task are evaluated by calculating the performance metrics such as accuracy, sensitivity, specificity, and positive predictive accuracy (PPA).

The accuracy versus number of epochs plots during training and testing of the proposed model are shown in Fig. 8 using different datasets. The losses during training and testing of the multi-task model with 30 numbers of epochs are shown in Fig. 9. In best-fit models, the validation loss is typically slightly higher than the training loss, with both learning curves being relatively close. The learning curves initially start with a somewhat high training and validation loss, which steadily decreases over time.

Additionally, the training and validation accuracy curves are quite similar across all datasets.

Fig. 7 displays the confusion matrix for the predictions related to the three datasets. The top-left, top-right, bottom-left, and bottom-right regions indicate true positive (TP), false positive (FP), false negative (FN), and true negative (TN) instances, respectively. All datasets exhibit the highest values for glaucoma images correctly predicted as glaucoma by the model. The second-highest values are for normal images correctly predicted as normal. The lowest values represent the classes that are incorrectly predicted. Therefore, the approach has demonstrated high performance across all three datasets. The quantitative result of the glaucoma detection task is shown in Table 5 which shows 97.43 % maximum accuracy by using the ORIGA dataset. The sensitivity, specificity, and PPA of 97.14 %, 97.77 %, and 98.07 % respectively are compared by using the other two datasets. The proposed model using REFUGE and DRISTI-GS datasets has also shown good results such as accuracy of 97.43 %, sensitivity of 97.14 %, specificity of 97.77 % and PPA of 98.07 % respectively.

Furthermore, the ROC (receiver operating characteristic) curve illustrates the classification performance at all thresholds, providing the trade-off between sensitivity and specificity. As shown in Fig. 10, all the ROC curves for the three datasets are closer to the top-left corner, indicating better performance. The proposed model is compared with two popular existing methods such as CNN and U-Net for both segmentation and classification tasks. The area under ROC (AUROC) is calculated using three datasets and it shows that using the ORIGA and DRISTI-GS datasets our model performs better than others which is 0.985.

The proposed glaucoma detection task is compared with some existing methods in Table 6. It is evident that, compared to other related studies employing classification techniques, our approach achieves higher accuracy values.

4.3. Discussion on achieved performance

The impressive performance metrics reported in the study, such as high dice coefficients and Jaccard indices for Optic Disc (OD) and Optic Cup (OC) segmentation, can be attributed to several key factors. The use of a modified U-Net architecture, which is known for its efficiency in image segmentation tasks, likely contributed significantly to the model's high accuracy. The U-Net's design allows for effective localization and precise boundary delineation, which is crucial for segmenting the OD and OC regions in retinal images. The inclusion of attention mechanisms, specifically designed to focus on Regions of Interest (ROI), enhances the model's ability to prioritize and accurately segment the critical areas of the fundus images. This targeted approach helps in capturing essential features, leading to better segmentation results. The use of GCN in the decoder part of the model architecture aids in capturing spatial relationships and enhancing the understanding of

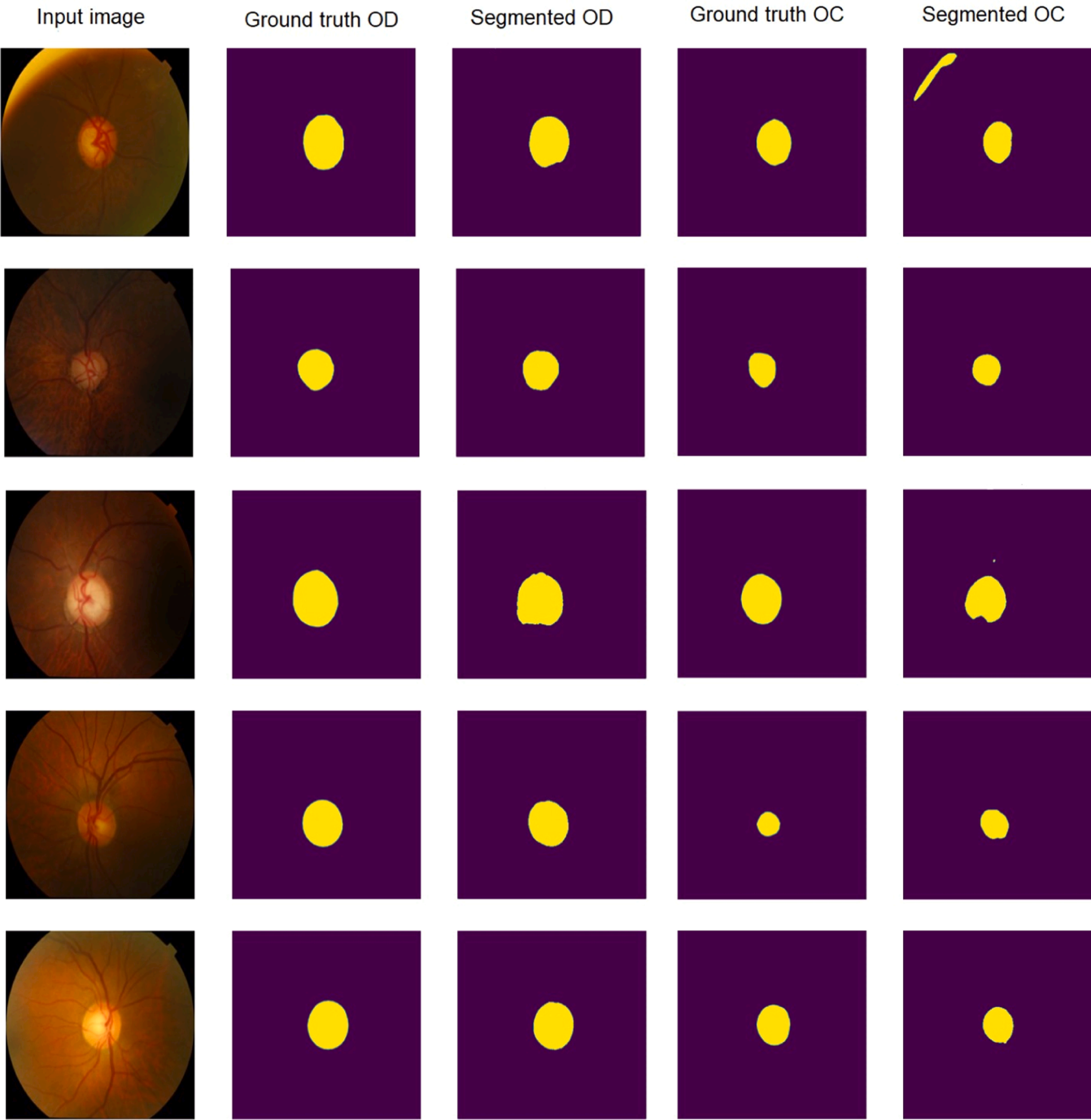


Fig. 6. Results of segmentation task compared with ground truth using three datasets.

Table 3
Evaluation parameters of segmentation task.

	OD	Segmentation	OC	Segmentation
Dataset	DC	Jc	DC	Jc
ORIGA	97.95	95.58	96.11	94.95
DRISTI-GS	96.56	91.27	94.11	93.14
REFUGE	96.59	92.87	95.23	92.95

structural features in the images. This capability is particularly useful for accurately segmenting complex and subtle regions like the OD and OC. Effective data augmentation techniques likely played a role in enhancing the model’s robustness and ability to generalize across

different datasets. Augmentation helps in creating a more diverse training set, thereby reducing the risk of overfitting and improving performance on unseen data. The model was trained and validated on three different datasets (ORIGA, DRISTI-GS, and REFUGE), each with unique characteristics. This variety helped the model to generalize better and perform consistently across different sets of images, as evidenced by the high dice and Jaccard coefficients across all datasets. The careful selection of training parameters, including the number of epochs and possibly other hyperparameters, contributed to achieving optimal model performance. Although not extensively detailed, effective tuning of these parameters is crucial for maximizing accuracy and minimizing errors. The availability of well-annotated data, which is essential for training supervised models, likely ensured that the model learned to

Table 4

Performance comparison with existing segmentation methods.

Author	Year	Technique	Dataset	OD Segmentation (%)	OC Segmentation (%)
Fu et al. [18]	2018	M-Net	DRISTI-GS, RIM-ONE	DC: 96.78, Jc: 93.86, DC: 95.26, Jc: 91.24	DC: 86.18, Jc: 77.3, DC: 83.48, Jc: 73
Son et al. [32]	2019	GAN	DRISTI-GS, RIM-ONE	DC: 95.27, Jc: 91.85, DC: 95.32, Jc: 91.22	DC: 86.43, Jc: 77.48, DC: 82.5, Jc: 71.65
Gu et al. [33]	2019	CE-Net	DRISTI-GS, RIM-ONE	DC: 96.42, Jc: 93.23, DC: 88.18, Jc: 80.06	DC: 95.27, Jc: 91.15, DC: 84.35, Jc: 74.24
Huang et al. [34]	2020	U-Net3+	RIM-ONEv3, DRISHTI-GS, REFUGE	DC: 95.5, 95.2, 95.9	DC: 84.3, 83.3, 83.7
Luo et al. [35]	2021	BGA-Net	RIM-ONEv3, DRISHTI-GS	DC: 96.7, 97.5	DC: 87.2, 89.8
Hu et al. [54]	2021	Encoder-decoder	REFUGE, MESSIDOR, RIM-ONE-r3, DRISHTI-GS, IDRiD	IoU: 0.9642, Dice:0.9314	IoU: 0.9003, Dice:0.8344
Fu et al. [55]	2021	U-Net	DRIVE, Kaggle, MESSI-DOR, and NIVE	GDOP: 1.12809	–
Pascal et al. [56]	2022	U-Net (VGG)	REFUGE	DSC: 0.954	DSC: 0.874
Proposed Model	–	Modified multitask U-Net	ORIGA, DRISTI-GS, REFUGE	DC: 97.95, Jc: 95.58, DC: 96.56, Jc: 91.17, DC: 96.59, Jc: 92.87	DC: 96.11, Jc: 94.95, DC: 94.11, Jc: 93.14, DC: 95.23, Jc: 92.95

* DC: Dice-coefficient, Jc: Jaccard, IoU: Intersection-Over-Union.

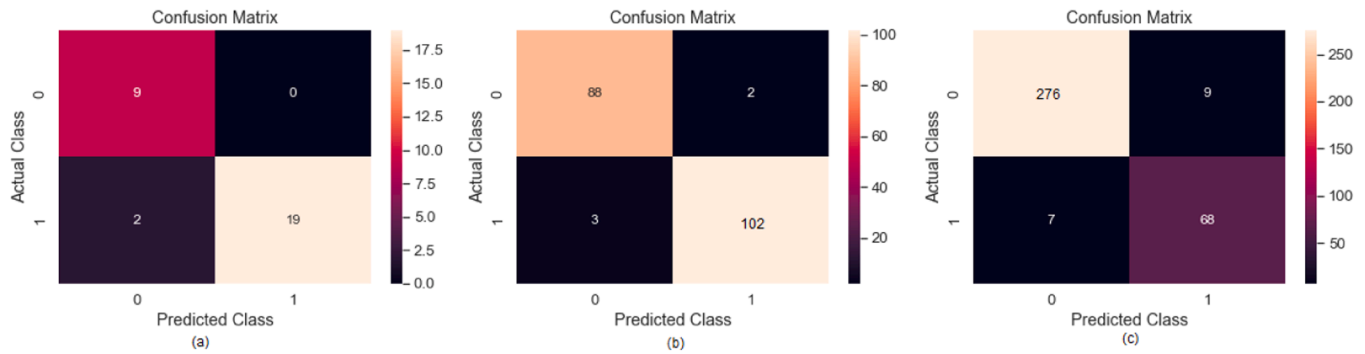
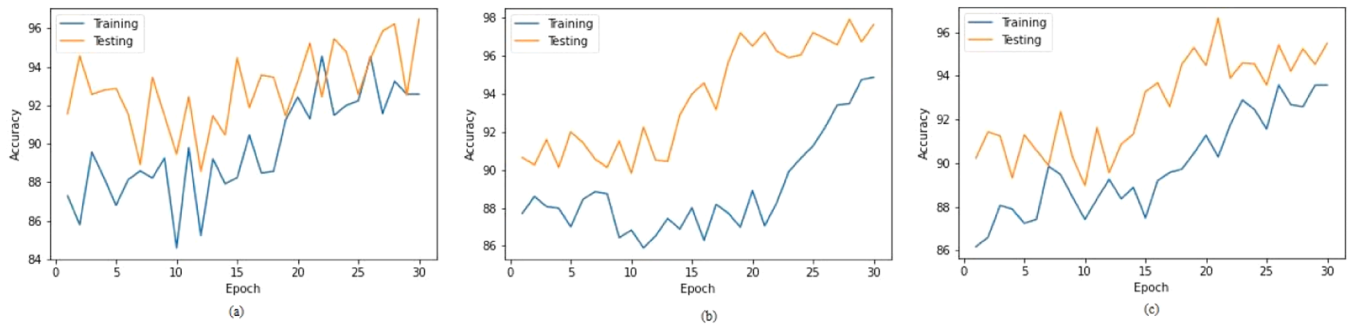
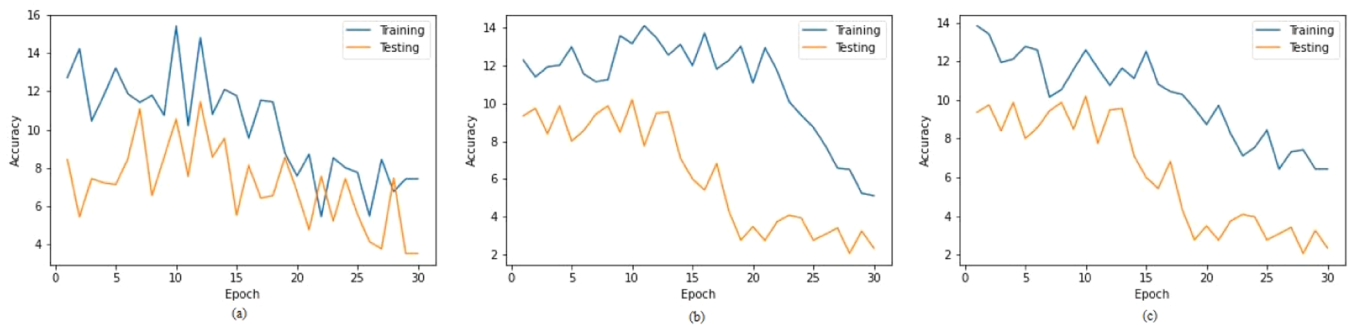
**Fig. 7.** Confusion matrix obtained after glaucoma classification using different datasets: (a) DRISTI-GS, (b) ORIGA, (c) REFUGE.**Fig. 8.** Accuracy plots during training and testing with no of epochs using different datasets: (a) DRISTI-GS, (b) ORIGA, (c) REFUGE.**Fig. 9.** Losses during training and testing with no of epochs using different datasets: (a) DRISTI-GS, (b) ORIGA, (c) REFUGE.

Table 5

Comparison of quantitative result on respective datasets for proposed model.

Dataset	Accuracy	Sensitivity	Specificity	Positive predictive accuracy
ORIGA	97.43	97.14	97.77	98.07
DRISTI-GS	93.33	90.47	100	100
REFUGE	95.55	90.66	96.84	88.31
Average	95.43	92.75	98.21	95.46

accurately identify and segment the relevant features in the retinal images. Overall, the combination of an advanced architecture, targeted feature extraction, diverse training data, and robust training processes enabled the model to achieve high performance in OD and OC segmentation across multiple datasets.

The high performance of the proposed multi-task deep learning model, as indicated by the accuracy, sensitivity, specificity, and Positive Predictive Accuracy (PPA) metrics, can be attributed to several critical factors. The use of a modified U-Net architecture, enhanced with attention mechanisms and a Graph Convolutional Network (GCN), provides a robust framework for extracting and learning from both global and localized features in the retinal fundus images. This architecture enables precise segmentation and effective classification, which are crucial for tasks such as glaucoma detection. The multi-task learning approach allows the model to share features between the segmentation and classification tasks. This not only reduces the overall complexity but also improves generalization by leveraging the interrelatedness of the tasks. This shared learning likely contributes to the high-performance metrics observed across various datasets. The consistent performance across different datasets, such as ORIGA, REFUGE, and DRISTI-GS, suggests that the model has been effectively trained and validated. The use of 30 epochs appears to strike a balance between underfitting and overfitting, as indicated by the accuracy and loss plots during training and testing. This careful calibration of training duration helps the model converge to a state that generalizes well to unseen data. The model's evaluation using metrics like sensitivity, specificity, and PPA provides a comprehensive view of its performance, especially in a clinical context where false negatives (missed cases) and false positives (misdiagnosed cases) can have significant implications. The high AUROC value of 0.985 indicates excellent discriminative ability of the model in distinguishing between glaucomatous and non-glaucomatous cases, further validated by the ROC curve comparisons. The model's performance was benchmarked against existing methods such as traditional CNNs and U-Net models. The fact that the proposed model outperforms these benchmarks, especially in the context of glaucoma detection and segmentation tasks, highlights the effectiveness of the architectural enhancements and multi-task approach. The inclusion of multiple datasets (ORIGA, REFUGE, and DRISTI-GS) for training and evaluation ensures that the model is tested across various image qualities and conditions. This diversity helps in building a more robust model that can handle different types of retinal images, thus improving the

generalizability of the results. Overall, the combination of advanced model architecture, careful training, comprehensive evaluation metrics, and rigorous comparative analysis contributes to the high accuracy and effectiveness of the proposed model in both segmentation and classification tasks. This performance demonstrates the model's potential for practical clinical applications in glaucoma detection.

When considering the limitations of this study, the model's effectiveness is contingent on comprehensive annotations, which are challenging and time-consuming to obtain. This reliance on detailed labeling may limit the model's generalization to datasets with less extensive annotations. The study primarily focuses on three specific datasets (ORIGA, REFUGE, and DRISTI-GS). The performance of the model on other datasets or real-world clinical data may vary. The use of a modified U-net architecture with Mobile-Netv2, Graph Convolution Network (GCN), and an Attention Module (AM) may increase the model's complexity, potentially affecting interpretability and ease of implementation. The study mentions training with 30 epochs, but does not explore the potential benefits of training for longer periods or fine-tuning other hyperparameters, which might further enhance performance. While the model shares features from a modified U-Net model, it may not fully capture all relevant features, particularly subtle ones that could be critical for accurate glaucoma detection and segmentation. With a relatively small dataset, there is a risk of overfitting, where the model performs well on the training data but fails to generalize to new,

Table 6

Performance comparison with existing classification methods.

Author	Year	Technique	Dataset	Performance
A. Prananda et al. [36]	2022	Alex-Net, Google-Net	ORIGA-light	Acc: 92.88 %, AUC: 89.34 %
R. Verma et al. [37]	2022	SVM, K-mean clustering, and convex hull	DRISHTIGS, real database	Acc: 85.39 %
J. Latif et al. [38]	2023	EGWO-SVM	ORIGA	ACC: 94 %, SEN: 92 %, SPE: 92 %
P. Yin et al. [58]	2022	DF-Net	DRIVE, STARE, CHASEDB1	ACC: 96.14 %, SEN: 97.04 %, SPE: 98.02 %
Gupta et al. [59]	2022	ODCNN-RFIC	ARIA, STARE	ACC: 89 %, SEN: 89.04 %
Alghamdi et al. [39]	2021	CNN with Autoencoder	RIM-ONE, RIGA	ACC: 93.8 %, SEN: 98.9 %, SPE: 90.5 %, AUC: 95 %
Joshi et al. [40]	2022	RES-Net, VGG-Net, Google-Net	DRIONS-DB, HRF, DRISHTIGS	ACC: 91.11 %, SEN: 85.55 %, SPE: 95.2 %
Proposed model	–	Modified U-Net multi-task	ORIGA, DRISTI-GS, REFUGE	ACC: 97.43 %, SEN: 97.14 %, SPE: 97.77 %, PPA: 98.07 %

* ACC: Accuracy, AUC: Area Under Curve, SEN: Sensitivity, SPE: Specificity, and PPA: Positive Predictive Accuracy.

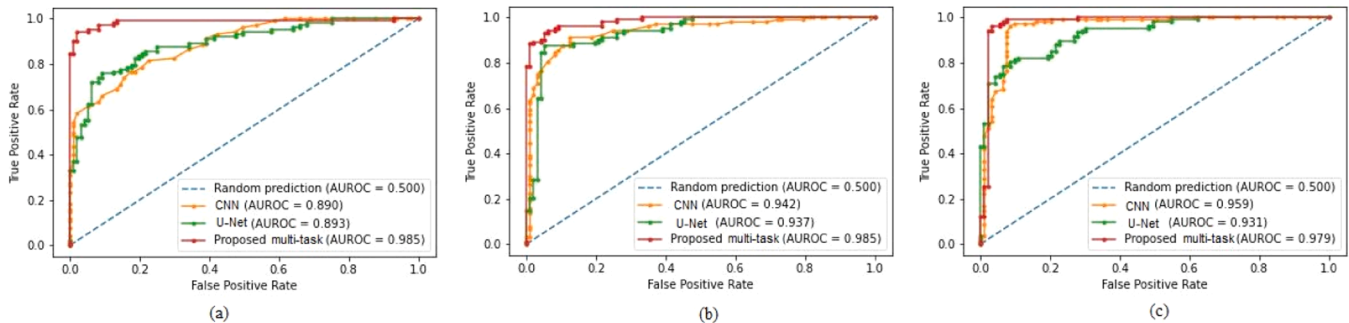


Fig. 10. ROC curve of proposed glaucoma detection method compared with other two existing methods for different datasets: (a) DRISTI-GS, (b) ORIGA, (c) REFUGE.

unseen data. The research assumes consistent image quality across the datasets used, which may not reflect real-world variability in image acquisition conditions. It is assumed that sharing features from the modified U-Net model across tasks reduces complexity and improves generalization, without significantly compromising task-specific performance.

5. Conclusion and future scope

In this research, a multi-task deep learning model is proposed to segment the optic disc, and optic cup, and to detect glaucoma using retinal fundus images. The features from the modified U-Net model are shared with all three multi-task systems to reduce the complexity and improve the generalization of the model. The proposed model is trained with 30 numbers of epochs and uses a smaller number of parameters as compared to other existing methods. The model is implemented using three datasets of a total of 1951 fundus images. The performance of the proposed model shows high-quality OD and OD segmented images with dice coefficients of 97.95 % and 96.11 % respectively. The classification task of the proposed model for accurate detection of glaucoma achieved better performance as compared to existing methods with a maximum accuracy of 97.43 %.

The model can be extended to include additional tasks such as vessel extraction, fovea localization, and detection of other ophthalmic diseases, potentially broadening its applicability. Future research could involve training and validating the model on more diverse datasets, including those with different image qualities and demographic variations, to enhance generalizability. Further exploration of optimal training parameters, data augmentation techniques, and feature extraction methods could improve the model's performance and efficiency. Employing explainable artificial intelligence frameworks can standardize the generation and presentation of explanations [57]. These frameworks can validate the model's performance and provide transparent insights into how decisions are made.

CRediT authorship contribution statement

Satyabrata Lenka: Writing – original draft, Validation, Software, Investigation, Formal analysis, Data curation. **Zefree Lazarus Mayaluri:** Writing – review & editing, Supervision, Methodology, Formal analysis, Conceptualization. **Ganapati Panda:** Writing – review & editing, Supervision, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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